

## CAPO CACCIA, Italy

WMO: 165220

Lat: 40.561N

Lon: 8.163E

Elev: 204

StdP: 98.90

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 4.6        | 5.9 | -1.9                          | 3.3 | 7.7  | 0.1 | 3.9 | 7.9  | 19.2                  | 10.5 | 16.9 | 9.9  | 6.3                   | 40   | 0.443 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 5.2                    | 31.9            | 21.1 | 30.1 | 21.2 | 28.8 | 21.6 | 25.5                | 27.6 | 24.6 | 26.8 | 23.9 | 26.2 | 3.5                  | 30   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 24.9                            | 20.5 | 26.9 | 24.0 | 19.3 | 26.2 | 23.1 | 18.3 | 25.5 | 79.5          | 27.6 | 75.8 | 26.9 | 72.5 | 26.2 | 30.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 15.3          | 13.2 | 11.4 | 2.5                        | 35.9 | 1.4                | 1.9 | 1.5                                                | 37.3 | 0.6        | 38.4 | -0.1       | 39.5 | -1.1       | 40.9 |    |    |
| (5)               |      |      | 0.9                        | 26.8 | 1.2                | 1.4 | 0.0                                                | 27.8 | -0.7       | 28.6 | -1.4       | 29.4 | -2.2       | 30.4 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 17.2  | 11.1  | 10.6  | 12.2  | 14.3  | 17.8  | 21.7  | 24.3  | 24.9  | 21.9  | 19.5  | 15.2  | 12.2 | (6)  |
| (7)  |                                                                   | DBStd       | 5.60  | 2.16  | 2.28  | 2.17  | 2.28  | 2.79  | 3.08  | 2.53  | 2.53  | 2.32  | 2.45  | 2.79  | 2.54 | (7)  |
| (8)  |                                                                   | HDD10.0     | 52    | 14    | 18    | 7     | 1     | 0     | 0     | 0     | 0     | 0     | 2     | 11    |      | (8)  |
| (9)  |                                                                   | HDD18.3     | 1107  | 226   | 215   | 190   | 124   | 43    | 4     | 0     | 0     | 2     | 16    | 97    | 191  | (9)  |
| (10) |                                                                   | CDD10.0     | 2676  | 47    | 36    | 76    | 128   | 243   | 352   | 444   | 462   | 357   | 294   | 159   | 78   | (10) |
| (11) |                                                                   | CDD18.3     | 690   | 0     | 0     | 0     | 2     | 27    | 107   | 186   | 204   | 109   | 52    | 4     | 0    | (11) |
| (12) |                                                                   | CDH23.3     | 3709  | 0     | 0     | 0     | 8     | 104   | 549   | 1191  | 1418  | 349   | 88    | 1     | 0    | (12) |
| (13) |                                                                   | CDH26.7     | 919   | 0     | 0     | 0     | 1     | 21    | 146   | 302   | 396   | 47    | 7     | 0     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 4.6   | 5.8   | 5.4   | 5.1   | 4.5   | 4.1   | 3.6   | 3.7   | 3.7   | 4.1   | 4.2   | 5.5   | 5.8  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 486   | 50    | 44    | 35    | 40    | 33    | 17    | 4     | 7     | 41    | 66    | 87    | 62   | (15) |
| (16) |                                                                   | PrecMax     | 783   | 145   | 145   | 79    | 113   | 111   | 65    | 28    | 40    | 112   | 198   | 234   | 151  | (16) |
| (17) |                                                                   | PrecMin     | 321   | 0     | 3     | 2     | 8     | 0     | 1     | 0     | 0     | 4     | 2     | 19    | 2    | (17) |
| (18) |                                                                   | PrecStd     | 115   | 32    | 33    | 20    | 26    | 29    | 17    | 7     | 8     | 30    | 46    | 48    | 33   | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 16.6  | 17.9  | 20.3  | 24.0  | 29.2  | 33.0  | 33.7  | 34.9  | 30.6  | 27.5  | 22.2  | 18.2 | (19) |
| (20) |                                                                   | MCWB        | 12.7  | 12.9  | 13.7  | 16.1  | 18.4  | 21.9  | 20.4  | 21.3  | 21.4  | 18.9  | 17.8  | 14.9  |      | (20) |
| (21) |                                                                   | 2%          | DB    | 15.2  | 15.1  | 17.7  | 21.0  | 25.9  | 30.1  | 31.2  | 32.1  | 27.8  | 25.5  | 21.0  | 17.0 | (21) |
| (22) |                                                                   | MCWB        | 12.8  | 12.0  | 13.1  | 14.7  | 17.4  | 20.5  | 20.8  | 21.1  | 21.5  | 19.2  | 17.2  | 14.7  |      | (22) |
| (23) |                                                                   | 5%          | DB    | 14.5  | 14.1  | 16.1  | 19.0  | 23.9  | 28.1  | 29.8  | 30.3  | 26.3  | 24.0  | 19.9  | 16.1 | (23) |
| (24) |                                                                   | MCWB        | 12.6  | 11.9  | 12.8  | 14.3  | 17.0  | 20.3  | 21.6  | 21.5  | 21.8  | 19.4  | 16.7  | 14.4  |      | (24) |
| (25) |                                                                   | 10%         | DB    | 13.9  | 13.3  | 15.0  | 17.5  | 22.0  | 26.3  | 28.1  | 28.8  | 25.2  | 22.8  | 18.9  | 15.3 | (25) |
| (26) |                                                                   | MCWB        | 12.2  | 11.4  | 12.6  | 14.1  | 16.7  | 20.5  | 22.0  | 22.1  | 21.5  | 19.2  | 16.2  | 13.7  |      | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 14.6  | 14.1  | 15.3  | 17.7  | 20.8  | 25.1  | 26.6  | 27.1  | 25.1  | 22.5  | 19.7  | 16.2 | (27) |
| (28) |                                                                   | MADB        | 15.2  | 15.6  | 17.4  | 21.3  | 23.6  | 27.5  | 28.7  | 28.5  | 26.5  | 24.1  | 21.2  | 17.1  |      | (28) |
| (29) |                                                                   | 2%          | WB    | 13.7  | 13.2  | 14.3  | 16.4  | 19.5  | 23.8  | 25.4  | 25.7  | 24.0  | 21.5  | 18.4  | 15.3 | (29) |
| (30) |                                                                   | MADB        | 14.5  | 14.2  | 16.1  | 18.9  | 22.6  | 26.3  | 27.7  | 27.6  | 25.5  | 23.3  | 19.9  | 16.4  |      | (30) |
| (31) |                                                                   | 5%          | WB    | 13.0  | 12.4  | 13.6  | 15.4  | 18.6  | 22.8  | 24.6  | 24.8  | 23.0  | 20.7  | 17.5  | 14.7 | (31) |
| (32) |                                                                   | MADB        | 14.0  | 13.6  | 15.2  | 17.6  | 21.7  | 25.5  | 26.9  | 26.9  | 25.3  | 22.7  | 19.1  | 15.8  |      | (32) |
| (33) |                                                                   | 10%         | WB    | 12.3  | 11.7  | 13.0  | 14.7  | 17.7  | 21.8  | 23.8  | 23.9  | 22.1  | 19.8  | 16.6  | 13.9 | (33) |
| (34) |                                                                   | MADB        | 13.5  | 13.1  | 14.5  | 16.7  | 20.7  | 24.9  | 26.2  | 26.4  | 24.4  | 21.9  | 18.3  | 15.2  |      | (34) |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 3.3   | 3.7   | 4.2   | 4.6   | 5.1   | 5.4   | 5.2   | 5.2   | 4.7   | 4.5   | 3.6   | 3.3   | (35) |      |
| (36) |                                                                   | 5% DB       | MCDBR | 3.6   | 4.6   | 5.9   | 7.2   | 8.4   | 8.6   | 8.1   | 8.1   | 6.3   | 6.2   | 4.5   | 3.7  | (36) |
| (37) |                                                                   | MCWBR       | 3.3   | 3.8   | 4.6   | 5.0   | 4.9   | 5.1   | 5.2   | 5.3   | 4.6   | 4.2   | 3.9   | 3.2   | (37) |      |
| (38) |                                                                   | 5% WB       | MADB  | 3.1   | 3.8   | 4.7   | 5.8   | 6.4   | 6.3   | 5.9   | 5.8   | 5.3   | 4.9   | 4.1   | 3.3  | (38) |
| (39) |                                                                   | MCWBR       | 3.4   | 3.8   | 4.2   | 4.8   | 4.8   | 5.3   | 5.3   | 5.2   | 4.6   | 4.1   | 4.1   | 3.3   | (39) |      |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.330 | 0.348 | 0.376 | 0.406 | 0.418 | 0.404 | 0.410 | 0.405 | 0.401 | 0.382 | 0.361 | 0.335 | (40) |      |
| (41) |                                                                   | taud        | 2.499 | 2.441 | 2.375 | 2.297 | 2.299 | 2.379 | 2.371 | 2.394 | 2.399 | 2.428 | 2.427 | 2.475 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 830   | 864   | 874   | 868   | 862   | 872   | 863   | 859   | 838   | 812   | 778   | 789   | (42) |      |
| (43) |                                                                   | Edh at Noon | 76    | 93    | 110   | 127   | 129   | 120   | 120   | 114   | 106   | 93    | 81    | 73    | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 1.90  | 2.79  | 4.22  | 5.44  | 6.54  | 7.58  | 7.64  | 6.59  | 4.97  | 3.48  | 2.15  | 1.70  | (44) |      |
| (45) |                                                                   | RadStd      | 0.17  | 0.28  | 0.37  | 0.37  | 0.54  | 0.26  | 0.25  | 0.34  | 0.18  | 0.27  | 0.16  | 0.17  | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |      |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |      |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     | (46) |
|      |                         | +0.44 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |      |
| (47) | Regional (13 neighbors) | +0.22 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     | (47) |

Nomenclature: See separate page